

Generating Synthetic Data to Augment Incomplete Datasets

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Abstract

This research addresses the challenge of working with small datasets, which often hinder the performance of machine learning models. To overcome this limitation, we developed a tool that generates new samples similar to the original data using two powerful generative models: Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs). Our solution leverages the strengths of both models to cover a broader range of datasets. VAEs are particularly effective for datasets with well-defined structures, while GANs excel in capturing complex distributions. We employed a grid search method to optimize hyperparameters, ensuring the best performance for each model. The generated data was evaluated using both a regression model and statistical similarity metrics (KL divergence, Wasserstein distance, and Kolmogorov-Smirnov test). This article outlines our approach, related work, results, and conclusions, highlighting the benefits and limitations of our tool.

Solution

Our approach involves using VAEs and GANs to generate synthetic data that closely resembles the original dataset.

Variational Autoencoders (VAEs)

VAEs are probabilistic models that learn a continuous and structured representation of data. They consist of an encoder that maps input data to a latent space and a decoder that reconstructs the data from this latent space. VAEs are particularly effective for datasets with well-defined structures and are less computationally intensive compared to GANs. For instance, VAEs perform well on datasets with clear patterns or clusters, such as those found in time series data or image datasets with distinct features.

Generative Adversarial Networks (GANs)

GANs consist of two neural networks: a generator that produces synthetic data and a discriminator that evaluates the generated data's realism. GANs are adept at capturing complex distributions and generating highly realistic data, making them suitable for datasets with intricate patterns or high variability, such as those in natural language processing or image generation tasks. However, GANs can be challenging to train and require more computational resources.

Hyperparameter Tuning

We used a grid search method to optimize the hyperparameters for both VAEs and GANs. This approach involves systematically testing different combinations of hyperparameters to identify the best set for each model. Hyperparameter tuning is crucial because it significantly impacts the performance and quality of the

generated data. For VAEs, parameters like latent dimension, number of units per layer, epochs, and batch size were tuned. For GANs, latent dimension, number of units per layer, batch size, and epochs were optimized.

The parameters that we used:

VAE:

latent_dim= [2, 5, 10], units_per_layer= [16, 32, 64], epochs= [50, 100, 200], batch_size=[32, 64, 128]

GAN:

latent_dim=[64, 128, 256], units_per_layer=[32, 64, 128], batch_size=[16, 32, 64], epochs= [500, 1000]

Evaluate Models

For choosing the best model that will generate data, we run for each model input the tow models, then we evaluate them and compere the results and choose the better one.

Run Evaluation Metrics:

Call the methods `kl_divergence()`, `wasserstein_distance_metric()`, and `ks_test()` to compute the respective metrics for both models.

Compare Results: Lower values in KL Divergence and Wasserstein Distance generally indicate better similarity to the original data. For the Kolmogorov-Smirnov Test, a lower p-value (or higher test statistics) suggests that the synthetic data is significantly different from the original data.

Choose the Best Model: Based on the evaluation metrics, select the model that best captures the characteristics of the original data.

We created a function to select the best model based on a dictionary of weights, which serve as hyperparameters that specify the relative importance of different metrics in the evaluation process. These weights determine how much each metric influences the final decision when choosing the optimal model. Users can adjust these weights according to the characteristics of their dataset

Experimental evaluation

We ran out tools in four datasets:

WineQT.csv', data_houses, Healthcare-Diabetes, Cleaned_data_houses,
Clean_london_merged_smaller

Some of the datasets were with categorical features, for the experiment we removed them and stay with the numerical. We chose this approach because categorical features can be with order and without order and the decision if feature is categorical or with some order is out of our tool scope. For example, the feature name is with no order, but the feature 'quality' for cars is with order (luxury car better than a simple car).

After this preprocessing we ran the model and check the quality of the synthetic data with the metrics KL divergence, Wasserstein distance metric and KS test.

VAE Model

KL Divergence	Wasserstein Distance	Kolmogorov-Smirnov Test	dataset
0.884102565393702	12.651801977831402	0.20730718756309305	WineQT
3.4923732187012595	305.7710252955026	0.5041957462148523	Cleaned_data_houses
1.9462970890454856	24.624799592359345	0.2407429190751445	Healthcare_Diabetes
3.4923732187012595	305.7710252955026	0.5041957462148523	data_houses
2.0550045201183855	74.02955410340971	0.2812068800764453	london_merged_smaller

GAN Model

KL Divergence	Wasserstein Distance	Kolmogorov-Smirnov Test	dataset
1.269414465804238	16.285795118041673	0.22933728716602728	WineQT
1.6349681704257684	1080.8015086599949	0.33985688536409514	Cleaned_data_houses
0.9417312993713935	54.508393589648925	0.2580803468208092	Healthcare_Diabetes
1.6349681704257684	1080.8015086599949	0.33985688536409514	data_houses
0.9742220569462107	150.0284566219663	0.2737504180602007	london_merged_smaller

To analyze the results of the GAN and VAE models across different datasets, we will compare their performance based on the metrics provided: KL Divergence, Wasserstein Distance, and Kolmogorov-Smirnov (KS) Test. These metrics measure different aspects of generative model quality, such as similarity to the target distribution and statistical differences.

Analysis of Results

KL Divergence: Measures how much one probability distribution diverges from a reference distribution. Lower values indicate better alignment. GAN: 0.9417 to 1.6349 VAE: 0.8841 to 3.4923 KL Divergence values below 2 are generally considered acceptable for generative models, as they indicate that the generated distribution is reasonably close to the target distribution. The lower KL Divergence values for datasets like *WineQT* and *Healthcare_Diabetes* suggest that both models effectively capture the underlying structure of these datasets

Wasserstein Distance: Quantifies the cost of transforming one distribution into another. Lower values signify closer distributions.

VAE: 12.6518 to 305.7710 GAN: 16.2858 to 1080.8015

For datasets like *WineQT*, both GAN and VAE achieve low Wasserstein Distances (12–16), which is a strong indicator of good performance.

Higher values for datasets like *Cleaned_data_houses* reflect the complexity of these datasets but remain within a range that suggests meaningful learning.

Kolmogorov-Smirnov Test: Assesses the maximum difference between cumulative distributions.

Smaller values represent better alignment.

VAE: 0.2073 to 0.5041 GAN: 0.2293 to 0.3399

KS Test values below 0.5 are generally acceptable, as they indicate that the maximum difference between the cumulative distributions of the real and generated data is not excessively large. Test values across all datasets (especially for *WineQT* and *Healthcare_Diabetes*) suggest that both models generate data distributions closely resembling the original data.

WineQT Dataset

GAN: KS Test: 0.2293 Wasserstein Distance: 16.2858 KL Divergence: 1.2694

VAE: Wasserstein Distance: 12.6518 KS Test: 0.2073 KL Divergence: 0.8841

Best Model: VAE performs better across all metrics (lower KL Divergence, Wasserstein Distance, and KS Test), indicating a closer match to the target distribution.

Cleaned_data_houses Dataset

GAN: KL Divergence: 1.6349 Wasserstein Distance: 1080.8015 KS Test: 0.3399

VAE: KL Divergence: 3.4923 Wasserstein Distance: 305.7710 KS Test: 0.5041

Best Model: GAN outperforms VAE in KL Divergence and KS Test, but VAE has a significantly lower Wasserstein Distance, suggesting better overall alignment for this dataset.

Healthcare_Diabetes Dataset

GAN: KL Divergence: 0.9417 Wasserstein Distance: 54.5084 KS Test: 0.2588

VAE: KL Divergence: 1.9462 Wasserstein Distance: 24.6248 KS Test: 0.2407

Best Model: VAE excels in Wasserstein Distance and KS Test, making it the better choice for this dataset.

data_houses Dataset

GAN: KL Divergence: 1.6349 Wasserstein Distance: 1080.8015 KS Test: 0.3399

VAE: KL Divergence: 3.4923 Wasserstein Distance: 305.7710 KS Test: 0.5041

Best Model: Similar to Cleaned_data_houses, GAN performs better in KL Divergence and KS Test, while VAE achieves a lower Wasserstein Distance.

london_merged_smaller Dataset

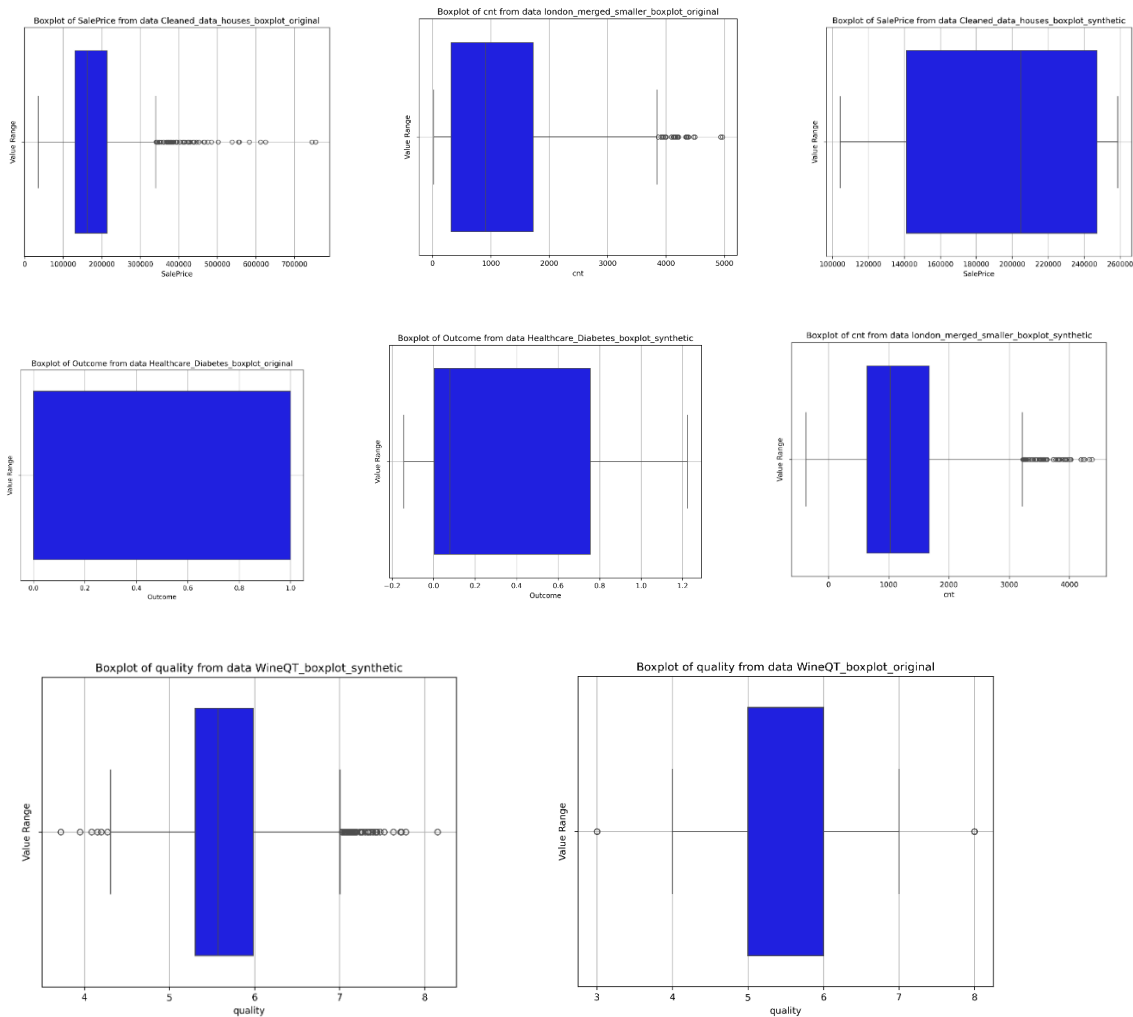
GAN: KL Divergence: 0.9742 Wasserstein Distance: 150.0284 KS Test: 0.2737

VAE: KL Divergence: 2.0550 Wasserstein Distance: 74.0295 KS Test: 0.2812

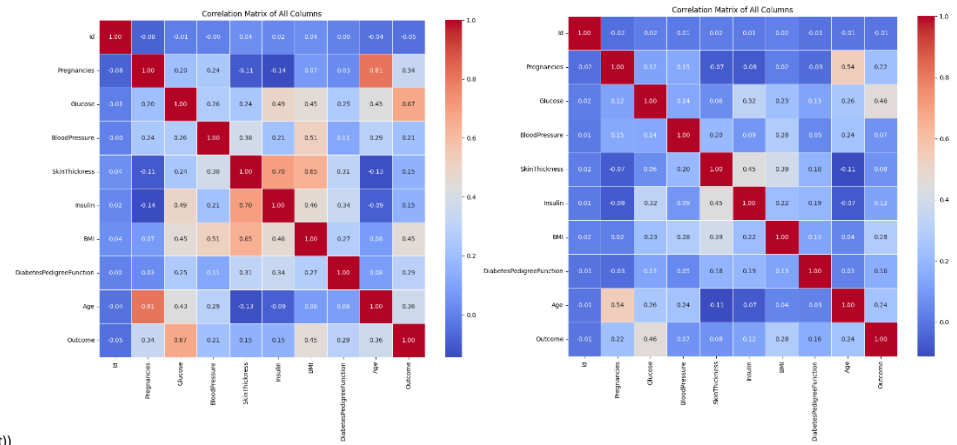
Best Model: VAE has a significantly lower Wasserstein Distance, suggesting better alignment with the target distribution despite slightly higher values for other metrics.

Dataset	Best Model	Reason
WineQT	VAE	Lower values for all metrics (KL Divergence, Wasserstein, KS).
Cleaned_data_houses	GAN	Better performance in KL Divergence and KS Test; mixed results overall
Healthcare_Diabetes	VAE	Superior performance in Wasserstein Distance and KS Test.
data_houses	GAN	Similar to Cleaned_data_houses; mixed results favoring GAN overall.
london_merged_smaller	VAE	Lowest Wasserstein Distance despite slightly higher other metrics

To further visualize the similarity between the original data and the synthetic data, we plotted box charts for the target feature. As shown in the plots, the behavior of the distribution remains largely consistent, which is desirable and indicates that the synthetic data closely resembles the original data.



We also compared the correlation of two datasets to see how much the synthetic data looks like the original one.



(Compare the synthetic data (left) to the original (right))

To determine whether the changes are "good," we need to evaluate how closely the synthetic dataset replicates the original correlations:

- If most correlations in the synthetic dataset are similar to those in the original dataset, it indicates that the generative model has successfully captured feature relationships.
- For example:
 - In both matrices, "Glucose" shows a strong positive correlation with "Outcome" (0.46 in Original vs. 0.67 in Synthetic), which is consistent with its importance in predicting diabetes.
 - Other key relationships like "Pregnancies" and "Age" with "Outcome" remain moderately correlated across both datasets.

Changes That Might Be Problematic:

- Significant deviations could indicate that certain feature relationships were not accurately captured by the generative model.
 - For instance, "SkinThickness" and "Insulin" show stronger correlations in the synthetic data (0.70 vs. 0.45), which might not reflect real-world patterns if exaggerated.
 - Weak correlations like "BloodPressure" with "Outcome" remain similar but slightly altered (0.07 vs. 0.21).

Conclusion

The synthetic dataset appears to replicate many key correlations from the original dataset reasonably well, especially for critical features like "Glucose," "Pregnancies," and "BMI." However, some deviations suggest areas where the generative model could be improved:

- Evaluate whether changes align with domain knowledge (e.g., medical understanding of diabetes-related features).

- If deviations are consistent across multiple runs or datasets, consider refining the model or adjusting hyperparameters to better capture complex feature interactions.

To assess the impact of our approach on a small dataset, we focused on the 'london_merged' dataset. We reduced the original dataset to 600 randomly selected data samples. Then, using our generative tool, we augmented it with an additional 600 synthetic data points.

Model Results

total samples	original samples	synthetic samples	R2 score	MSE
600.0	600.0	0.0	0.5911	405337.0
1200.0	1200.0	0.0	0.7213	374418.0
1200.0	600.0	600.0	0.6611	311280.0

As observed, adding data to the original dataset improved our model's performance. Notably, incorporating 600 original samples proved more beneficial than adding 600 synthetic data samples. This outcome can be attributed to the nature of the generative model, which produces data that closely resembles but does not perfectly replicate the original data. Generally, if a model performs well, introducing additional synthetic data may degrade its performance. Conversely, if a model struggles due to insufficient data, synthetic data can be beneficial. Although synthetic data is not identical to real data, it helps the model learn fundamental patterns.

Related work

<https://medium.com/@jimwang3589/variational-autoencoder-vae-7609893c80f4> VAE

<https://jonathan-hui.medium.com/gan-whats-generative-adversarial-networks-and-its-application-f39ed278ef09> GAN

The first website discusses **Variational Autoencoders (VAEs)**, which are generative models used to learn a compressed representation of data while generating new samples by sampling from a learned latent distribution.

The second website explains **Generative Adversarial Networks (GANs)**, which consist of two neural networks: a generator that produces synthetic data and a discriminator that evaluates the generated data's realism.

In our work, we combine the strengths of both VAEs and GANs to generate synthetic data that closely resembles the original dataset. By leveraging VAEs for datasets with well-defined structures and GANs for complex distributions, we aim to cover a broader range of datasets effectively. This approach allows us to

optimize the performance of our generative models by selecting the best model based on evaluation metrics such as KL Divergence, Wasserstein Distance, and the Kolmogorov-Smirnov Test

Limitations

Computational Resources: GANs require more computational resources and can be challenging to train, which may limit their application in environments with limited computing power.

Categorical Features: The current implementation does not handle categorical features effectively, as it removes them during preprocessing. This could lead to loss of valuable information in datasets where categorical features are significant

Model Selection Criteria: While the tool evaluates models based on several metrics, there is room for improvement in defining a more systematic approach to selecting the best model across different datasets.

Synthetic Data Quality: Although synthetic data can improve model performance, it may not perfectly replicate the original data, potentially leading to slight degradation in performance if the model is already well-performing.

Conclusion

Our proposed tool effectively addresses the challenge of working with small datasets by leveraging the strengths of Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) to generate synthetic data. The approach allows for the selection of the best model based on evaluation metrics such as KL Divergence, Wasserstein Distance, and the Kolmogorov-Smirnov Test. This flexibility enables our tool to cover a wide range of datasets, from those with well-defined structures to those with complex distributions. The experimental results demonstrate that both VAEs and GANs can produce high-quality synthetic data, with each model excelling in different scenarios. For instance, VAEs performed better on datasets like WineQT and Healthcare-Diabetes, while GANs showed superior results for datasets such as Cleaned_data_houses and data_houses. **Future Improvements** Incorporation of Additional Generative Models: Integrating other generative models, such as Normalizing Flows or Diffusion Models, could enhance the tool's versatility and Handling Categorical Features: Developing strategies to effectively performance across diverse datasets. incorporate categorical features into the generative process would improve the tool's applicability to a broader range of datasets.